

Taxation and the Welfare State: Solutions to Problem Set 8

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Question 1)

a./b.

Branches → economic risk:

- Unemployment insurance → becoming unemployed
- Health insurance → healthcare expenditure
- Old-age pensions → (i) longevity risk, (ii) not being able to work at old age
- Disability insurance → not being able to work due to disability
- Long-term care insurance → not being able to care for oneself at old age
- Accident insurance → healthcare expenditure/lost earnings due to work accident

c.

Numbers for 2024:

- Old-age pensions: $\approx$ €400bn
- Health insurance: €327bn
- Unemployment insurance €44bn
- Disability insurance: included in pensions; expenditure on disability benefits $\approx$ €21bn
- Long-term care insurance: €68bn
- Accident insurance: €16bn

d.

Reducing expenditure on old-age pensions and health insurance would be the biggest reforms in terms of potential savings. These two account for the majority of social insurance expenditure and are growing over time due to population aging. Another program that is growing very fast is long-term care insurance – if the government is worried about fiscal sustainability this would be an area to consider. For a full normative analysis of whether and where the government *should* cut expenditure on social insurance programs, we would need to consider both the insurance value and the fiscal/incentive cost of each program (see the Baily-Chetty model in Lecture 10).

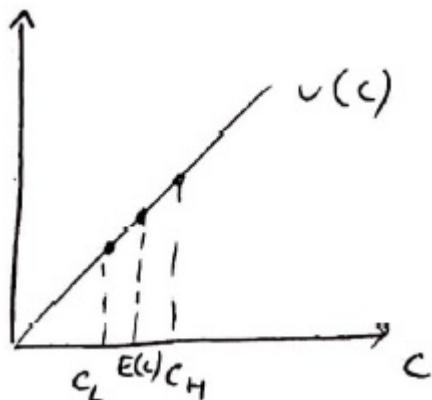
Question 2)

a.

[see Lecture 9, slides 15-16]

b.

Risk-neutral agents have no reason to buy insurance. Risk-neutral preferences are given by $u'(c) > 0$, $u''(c) = 0$ as shown in the graph below. For a risk-neutral individual, $Eu(c) = u(E(c))$, and hence the individual would see no utility gain from smoothing consumption across states of the world.



Question 3)

a.

If risk types are observable, the insurer will offer each type full coverage at its actuarially fair premium (€20 / €200 / €400 / €800 / €1,600). Since everyone is risk-averse, each strictly prefers full insurance at a fair premium to bearing the risk, so all buy. This is the efficient outcome: every risk-averse individual is fully insured.

b.

Initial premium = population-average actuarially fair premium, €278. A type buys only if the premium is worth it. The lowest-risk types A and B likely do not buy the insurance. [Note: since all types are risk-averse, their willingness to pay for insurance $\geq$ the expected damage. So technically, types A and especially B may want to buy the insurance at the average price if they are very risk-averse. We focus here on the case where they are not willing to buy at this initial price.]

c.

Iterate. With only types C, D, E in the pool (shares 0.25/0.05/0.05), average risk = $0.11/0.35 = 0.314 \rightarrow$ premium = €629. Now type C exits. The insurer offers a pooled price for types D, E: average risk = $0.06/0.10 = 0.60 \rightarrow$ premium = €1,200. Now D exits. E remains, their risk is $p = 0.80 \rightarrow$ premium = €1,600, which is exactly E's fair premium. Only type E (very poor health) ends up insured. This is inefficient: four risk-averse types A to D who would all be insured under full information are now priced out, preventing mutually beneficial trades.

d.

A mandate that puts everyone back in the pool at the uniform premium €278 restores full insurance for all. Efficiency improves relative to (c) because the four risk-averse types are insured again, i.e. adverse selection is eliminated. But not everyone is better off than in (c): the low-risk types subsidize the high-risk. Types C, D, E gain (they pay €278 against fair premiums of €400/€800/€1,600). A and B are worse off because they have to pay more than their willingness to pay for the insurance.

e.

The market outcome charges the sickest €1,600 and the healthiest €20 for the same coverage. If health risk is largely an "endowment", reflecting genetics or bad luck, the social planner might judge it unfair to make people pay for their individual risk. Social insurance with income-based contributions redistributes from low-risk to high-risk (and from high- to low-income). [A potential counterargument is that risk differences may partly reflect choices, such as lifestyle, smoking, etc., which society may not want subsidize via social insurance.]